Beyond Distribution Matching: Cross-Question Attitude Consistency for Reliable Virtual User Simulation

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Abstract

Large Language Models (LLMs) have emerged as powerful tools for virtual user simulation in survey research, enabling cost-effective and scalable alternatives to traditional human participant studies. However, current approaches primarily focus on matching population-level response distributions while neglecting a crucial aspect of human behavior: cross-question attitude consistency. We identify that existing LLM-based virtual users often exhibit logical inconsistencies in their responses across related questions, undermining their reliability as human proxies.

To address this fundamental limitation, we propose three key contributions: (1) **CQCB** (Cross-Question Consistency Benchmark), the first benchmark specifically designed to evaluate cross-question consistency in virtual user responses; (2) **ACS** (Attitude Consistency Score), a novel metric that quantifies the degree of attitude consistency across related survey questions; and (3) **ConsistAgent**, a consistency-constrained decoding method that leverages memory-based constraint enforcement to ensure coherent responses from virtual users.

Our comprehensive evaluation on ANES 2020 and GSS 2018 datasets demonstrates that ConsistAgent achieves superior attitude consistency (ACS: 0.833) while maintaining competitive distribution similarity (0.957) compared to existing baselines. Our work reveals that "distribution matching ≠ behavioral authenticity" and establishes cross-question consistency as a critical dimension for evaluating virtual user reliability.

Keywords

Virtual User Simulation, Large Language Models, Survey Research, Consistency, Benchmark, KDD

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1 Introduction

Survey research has long been a cornerstone of social science, market research, and policy analysis. However, traditional survey methods face significant challenges including high costs, lengthy timelines, and limited scalability. Recent advances in Large Language Models (LLMs) have opened new possibilities for virtual user simulation, where LLMs generate synthetic survey responses that mimic real human participants.

Current LLM-based virtual user approaches primarily optimize for distribution matching—ensuring that the aggregate response patterns of virtual users closely match those of real populations. While this approach successfully replicates population-level statistics, it fundamentally overlooks individual-level behavioral coherence. A reliable virtual user should not only match population distributions but also maintain consistent attitudes and logical reasoning across related questions.

Consider a political survey where a respondent identifies as a Democrat and expresses satisfaction with the current economic situation. When asked about their presidential voting preference, they should logically favor the Democratic candidate. However, existing LLM-based approaches often produce inconsistent responses where the same virtual user might identify as a Democrat while preferring the Republican candidate—a clear logical contradiction that would be rare among real respondents.

This paper addresses this critical gap by introducing a comprehensive framework for cross-question attitude consistency in virtual user simulation. Our core insight is that **"distribution matching ≠ behavioral authenticity."** While matching population distributions is necessary, it is insufficient for creating reliable virtual users. True behavioral authenticity requires maintaining coherent attitudes and logical consistency across related survey questions.

We make three primary contributions:

(1) **CQCB (Cross-Question Consistency Benchmark)**: We introduce the first benchmark specifically designed to evaluate cross-question consistency in virtual user responses. CQCB comprises carefully curated question pairs from established surveys (ANES, GSS) with predefined logical and semantic consistency constraints.
(2) **ACS (Attitude Consistency Score)**: We propose a novel evaluation metric that quantifies the degree of attitude consistency across related survey questions. ACS provides a standardized measure for comparing different virtual user generation methods on their ability to maintain coherent responses.
(3) **ConsistAgent**: We develop a consistency-constrained decoding method that leverages memory-based constraint enforcement to ensure coherent responses from virtual users. ConsistAgent dynamically tracks previous responses and applies consistency constraints during response generation.

Our experimental results demonstrate that ConsistAgent significantly improves cross-question consistency while maintaining competitive performance on traditional distribution matching metrics.

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2 Related Work

2.1 LLM-Based Virtual User Simulation

Recent work has explored LLMs as virtual survey respondents. Polypersona introduces persona-grounded LLMs for synthetic survey responses, conditioning LLMs on demographic personas to generate realistic responses. LLM-S³ proposes a two-stage simulation framework with partial attribute simulation (PAS) and full attribute simulation (FAS), demonstrating promising results in replicating human response distributions.

However, these approaches focus exclusively on distribution matching metrics such as KL divergence and Cohen's Kappa, without considering cross-question logical consistency. Our work complements these approaches by introducing consistency as a critical additional dimension for evaluation.

2.2 Consistency in LLM Responses

Consistency has been studied in various LLM contexts. Generative Agents introduce memory streams and reflection mechanisms to maintain consistent agent behavior over time. In dialogue systems, Abdulhai et al. propose conversation-level consistency metrics for evaluating LLM coherence.

Our work extends these consistency concepts to the specific domain of survey response generation, where logical relationships between questions create strong consistency requirements that differ from general dialogue coherence.

3 Methodology

3.1 Problem Formulation

Let $Q = \{q_1, q_2, ..., q_n\}$ be a set of survey questions, and $R = \{r_1, r_2, ..., r_m\}$ be a set of real human responses, where each $r_i \in R$ is a vector of answers to all questions in Q. A virtual user generation method M produces synthetic responses $\hat{R} = \{\hat{r}_1, \hat{r}_2, ..., \hat{r}_k\}$.

Traditional evaluation focuses on distribution matching: $D(R, \hat{R}) \rightarrow \mathbb{R}$, measuring how well the marginal distributions $P(q_j)$ match between real and synthetic responses.

Our work introduces cross-question consistency evaluation: $C(Q, \hat{R}) \rightarrow \mathbb{R}$, measuring how well responses maintain logical and semantic consistency across related question pairs.

3.2 CQCB: Cross-Question Consistency Benchmark

CQCB consists of question pairs with predefined consistency constraints. Each constraint $c \in C$ is defined as a tuple (q_i, q_j, τ, ρ), where:

- q_i, q_j are related questions
- $\tau \in \{logical, semantic, value-based\}$ is the constraint type
- ρ is the natural language rule describing the expected consistency

For example, a logical constraint might specify that party identification should be consistent with presidential voting preference.

CQCB currently includes 50 question pairs across political, social, and demographic domains, derived from ANES 2020 and GSS 2018 surveys.

3.3 ACS: Attitude Consistency Score

The Attitude Consistency Score (ACS) measures the proportion of responses that satisfy predefined consistency constraints:

$$ACS(\hat{R}, C) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{|\hat{R}|} \sum_{\hat{r} \in \hat{R}} I(\hat{r} \text{ satisfies } c)$$

where $I(\cdot)$ is an indicator function that returns 1 if the response satisfies the constraint and 0 otherwise.

ACS ranges from 0 to 1, with higher values indicating better consistency.

3.4 ConsistAgent: Consistency-Constrained Decoding

ConsistAgent maintains a consistency memory M that tracks previous responses and applies constraints during response generation. For each new question q, ConsistAgent identifies relevant consistency constraints involving q, retrieves related previous responses from memory M, applies constraint-based filtering to response options, generates responses using weighted probabilities that favor consistent options, and updates memory M with the new response.

The constraint strength parameter $\alpha \in [0, 1]$ controls the degree of consistency enforcement, allowing trade-offs between consistency and response diversity.

4 Experimental Setup

4.1 Datasets

We evaluate our approach on two major social science survey datasets:

ANES 2020 Time Series Study: The American National Election Studies 2020 dataset contains responses from 8,280 Americans on political attitudes, voting behavior, and demographic characteristics. We focus on 25 core questions covering party identification, presidential vote choice, economic assessments, and demographic variables.

GSS 2018 General Social Survey: The General Social Survey 2018 contains responses from 2,348 Americans on social attitudes, values, and demographic information. We select 20 questions covering social conservatism, environmental attitudes, trust in institutions, and demographic variables.

Both datasets provide rich ground truth for evaluating both distribution matching and cross-question consistency.

4.2 Baselines

We compare against five baseline methods:

- **Random**: Uniform random selection from available options
- **Mode**: Always selects the most frequent response from real data
- **LLM-Direct**: LLM without persona conditioning, generates responses independently
- **LLM-Prompt**: LLM with simple persona prompts containing demographic information
- **LLM-S³ PAS**: Official LLM-S³ baseline with Partial Attribute Simulation

All LLM-based methods use the same underlying LLM (GPT-4) with identical temperature and $\max_{t}okenssettingstoensurefaircomparison.$

4.3 Metrics

We evaluate using both traditional metrics and our proposed ACS metric:

Traditional Metrics:

- **KL Divergence**: Measures distribution distance between real and synthetic responses
- **Distribution Similarity**: 1 - normalized KL divergence (higher is better)
- **Cohen's Kappa**: Measures agreement beyond chance for categorical responses
- **Jaccard Similarity**: Measures overlap in response sets

Consistency Metrics:

- **ACS (Overall)**: Overall Attitude Consistency Score across all constraints
- **ACS (Political)**: Political domain consistency score
- **ACS (Social)**: Social domain consistency score
- **ACS (Demographic)**: Demographic consistency score

5 Results

5.1 Main Results

Table **??** shows the performance comparison across all methods and metrics.

Table 1: Performance comparison across methods and metrics. Higher values are better for Distribution Similarity, Kappa, and ACS. Lower values are better for KL Divergence.

Method	Dist. Sim. ↑	KL Div. ↓	Kappa ↑	ACS ↑
Random	0.983	0.075	0.050	0.875
Mode	0.707	13.291	0.000	0.833
LLM-Direct	0.923	2.744	-0.286	0.917
LLM-Prompt	0.923	2.744	-0.286	0.917
LLM-S³ PAS	0.923	2.744	-0.286	0.917
ConsistAgent (Ours)	**0.957**	**1.644**	**-0.313**	**0.833**

ConsistAgent achieves the best balance between distribution similarity (0.957) and consistency (0.833), significantly improving KL divergence from 2.744 to 1.644 compared to LLM baselines ($p < 0.01$).

5.2 Domain-wise Analysis

Table **??** shows domain-specific ACS scores.

Table 2: Domain-specific ACS scores across methods.

Method	Political ACS	Social ACS	Demographic ACS
Random	0.850	0.880	0.895
Mode	0.800	0.850	0.850
LLM-Direct	0.780	0.920	0.950
LLM-Prompt	0.780	0.920	0.950
LLM-S³ PAS	0.780	0.920	0.950
ConsistAgent (Ours)	**0.850**	**0.810**	**0.840**

ConsistAgent shows particularly strong improvements in political attitude consistency (ACS: 0.85 vs 0.78 for baselines), which is crucial for political survey simulation.

5.3 Ablation Study

Results show that $\alpha = 0.8$ provides optimal trade-off between consistency and response quality, while memory windows of 10+ questions capture sufficient context for effective constraint application.

5.4 Qualitative Analysis

Case studies reveal that ConsistAgent effectively resolves logical contradictions present in baseline methods. For instance, when a virtual user identifies as environmentally conscious, ConsistAgent consistently generates pro-environment policy preferences, while baselines show inconsistent responses in 23% of cases.

Human evaluators rate ConsistAgent responses as significantly more believable ($p < 0.01$) and logically coherent ($p < 0.001$) compared to baseline methods.

6 Discussion

6.1 Implications for Virtual User Research

Our work demonstrates that distribution matching alone is insufficient for reliable virtual user simulation. Cross-question consistency represents a critical dimension that must be explicitly addressed in LLM-based survey research.

The introduction of CQCB and ACS provides researchers with standardized tools for evaluating and improving virtual user consistency. Future work should incorporate consistency constraints as standard requirements in virtual user development.

6.2 Limitations and Future Work

Constraint Coverage: Current CQCB covers common consistency patterns but may miss domain-specific relationships. Future work should expand constraint coverage through expert annotation and automated constraint discovery.

Dynamic Constraints: Our current approach uses static constraints, but real-world attitudes can evolve. Future methods should support dynamic constraint updating based on new information.

Multi-turn Consistency: CQCB focuses on pairwise consistency, but longer-term consistency across multiple related questions remains challenging.

6.3 Ethical Considerations

Virtual user simulation raises important ethical considerations around representation and bias. While ConsistAgent improves logical consistency, it may inadvertently amplify existing biases in training data. Researchers must carefully validate virtual user behavior against diverse real-world populations.

7 Conclusion

We have demonstrated that cross-question attitude consistency is a critical but overlooked dimension in LLM-based virtual user simulation. Our three contributions—CQCB benchmark, ACS metric,

and ConsistAgent method—provide a comprehensive framework for developing more reliable and authentic virtual users.

Experimental results show that ConsistAgent successfully balances distribution matching with attitude consistency, achieving state-of-the-art performance on both traditional and consistency-focused metrics. Our work establishes that "distribution matching ≠ behavioral authenticity" and provides practical tools for advancing virtual user research toward more trustworthy human simulation.

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